AI-Face: A Million-Scale Demographically Annotated AI-Generated Face Dataset and Fairness Benchmark*

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Abstract

The rapid proliferation of AI-generated faces—spanning deepfakes, GANs, and diffusion models—poses serious risks to media integrity, yet current detectors exhibit significant performance disparities across demographic groups. Existing datasets either lack demographic labels or cover only a narrow range of generative methods, hindering the development and fair evaluation of detection systems. We introduce ***AI-Face****, the first million-scale, demographically annotated AI-generated face dataset and fairness benchmark. AI-Face comprises 1,646,545 face images (400,885 real; 1,245,660 fake) spanning 37 generation methods across deepfake videos, GANs, and diffusion models. Each image is annotated with skin tone (using the Monk Skin Tone scale), gender, and age via a novel CLIP-based annotator and colorimetric skin-tone estimator. Using AI-Face, we benchmark 12 detectors from four families across utility metrics (AUC, ACC, AP, EER, FPR), group and intersectional fairness metrics (F_{DP}, F_{MEO}, F_{EO}, F_{OAE}, F_{IND}), reliability under six post-processings, and cross-domain generalization. Our analyses reveal systematic bias patterns—female and light-skin subgroups suffer higher false positive rates, and diffusion-model subsets induce the largest fairness gaps. AI-Face establishes a rigorous, scalable foundation for building equitable and reliable face forensics systems.*

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1. Introduction

The emergence of highly realistic AI-generated faces has dramatically lowered the barrier to creating deceptive media. From deepfake videos [1–3] to GAN-synthesized portraits [4–6] and diffusion-model imagery [7–9], the landscape of synthetic face generation is both expansive and rapidly evolving [10]. This proliferation has fueled real-world harms including political disinformation [11, 12], non-consensual synthetic media [13], and identity fraud. Consequently, the detection of AI-generated faces has attracted considerable research attention [14–17].

Despite progress in detection accuracy, a critical dimension has been largely overlooked: *fairness*. Recent audits suggest that detectors systematically misclassify faces from certain demographic groups at higher rates, raising ethical and legal concerns [18, 19]. However, rigorously studying and mitigating such biases requires a large-scale dataset with high-quality demographic annotations spanning diverse generative methods—a resource that does not yet exist.

Limitations of existing resources. As summarized in Table 1, existing demographically annotated face datasets cover at most a few generation methods and typically omit one or more of the three key attributes (skin tone, gender, age). Concurrent benchmark efforts such as DeepfakeBench [14], DF40 [17], and CDDB [20] evaluate utility but omit systematic fairness analysis (Table 3). Those that do assess fairness [18, 19] are restricted to deepfake videos and do not cover GANs or diffusion models, nor do they measure reliability of fairness under distribution shift.

Our contribution: AI-Face. We present AI-Face, a comprehensive benchmark designed from the ground up for fairness-aware evaluation of AI-generated face detectors. AI-Face makes the following contributions:

- **Million-scale, method-diverse dataset**: 1,646,545 face images spanning 37 generation methods—including 4 deepfake video pipelines, 10 GAN variants, and 8 diffusion models—with real images sourced from FFHQ [4], IMDB-WIKI [21], FF++ [1], DFDC [2], DFD [22], and Celeb-DF-v2 [3].
- **High-quality demographic annotations**: Skin tone is measured using a colorimetric pipeline (MediaPipe FaceMesh [23] → K-Means clustering [24] → Monk Skin Tone scale [25, 26] mapping). Gender and age are predicted by a CLIP ViT-L/14 [27] backbone with a 3-layer MLP, trained with Vector Scaling loss [28] and a Sinkhorn-based fairness regularizer [29], and optimized with SAM [30]. A three-person user study validates annotation quality, showing AI-Face substantially outperforms prior annotated datasets.
- **Comprehensive fairness benchmark**: We evaluate 12 detectors spanning Naive, Frequency, Spatial, and Fairness-enhanced families on group fairness (F_{DP}, F_{MEO}, F_{EO}, F_{OAE}) and individual fairness (F_{IND}), as well as intersectional fairness across skin-tone×gender subgroups. We

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further assess robustness under six post-processing transformations and generalization to three external datasets (CCv2 [31], DF-Platter [32], GenData [33]).

- **Actionable insights**: Our analyses reveal that female and light-skin subgroups experience disproportionately high false positive rates, diffusion-model subsets induce the largest fairness gaps, and fairness-enhanced methods do not consistently dominate standard detectors across all metrics.

AI-Face thus provides a rigorous, scalable foundation for developing equitable AI-generated face detectors. We publicly release the dataset, annotations, and benchmark code.

2. Related Work

AI-generated face detection. Early detectors focused on deepfake video artifacts [1–3] using naive CNN classifiers such as Xception [34] and EfficientNet [35]. Frequency-domain methods [36–38] exploit spectral cues left by generative processes. Spatial methods [39–41] leverage richer visual representations for broader generalization. Fairness-enhanced detectors [42, 43] explicitly incorporate demographic priors. Comprehensive benchmarks including DeepfakeBench [14], DF40 [17], CDDB [20], and the survey of Le et al. [15] evaluate utility across diverse conditions, but do not systematically assess demographic fairness.

Demographically annotated face datasets. Several datasets provide demographic labels alongside AI-generated content. DeePhy [44] and DF-Platter [32] include skin tone, gender, and age but cover only 3 generation methods. The annotated collections of Xu et al. [45] (A-FF++, A-DFDC, A-DFD, A-Celeb-DF-v2, A-DF-1.0) cover deepfake videos exclusively. GenData [33] evaluates bias but covers only GAN and diffusion methods without deepfake videos. As shown in Table 1, no existing dataset combines million-scale coverage with all three generation categories and all three demographic attributes.

Fairness in face analysis. Algorithmic fairness has been extensively studied in face recognition [46, 47], where disparities along race, gender, and age axes are well-documented [48]. Studies of face detection [18] and video forensics [19] have identified similar disparities. Our work builds on these findings and provides the first rigorous fairness benchmark that jointly covers generation diversity, demographic annotations, and reliability testing for AI-generated face detectors.

Generative models. AI-Face samples from a broad generative landscape: deepfake video methods (FaceSwap, FaceShifter, DeepFakes, etc.), GAN variants including StyleGAN [4], StyleGAN2 [5], StyleGAN3 [49], BigGAN [6], VQGAN [50], StarGAN/STGAN [51], ProGAN [52], and diffusion models including Stable Diffusion [7], Midjourney [53], DALL-E 2 [8], and DeepFloyd IF [9].

3. AI-Face Dataset

3.1. Data Collection and Scale

AI-Face contains **1,646,545** face images organized into 28 subsets: 400,885 real images and 1,245,660 AI-generated images across 37 generation methods. Figure 1 illustrates how AI-Face compares with prior datasets in demographic annotation coverage, generation category diversity, and the number of generation methods covered.

Real images are sourced from FFHQ [4] and IMDB-WIKI [21] for their demographic diversity, plus the real partitions of FF++ [1], DFDC [2], DFD [22], and Celeb-DF-v2 [3].

Fake images span three generation categories: (1) *Deepfake videos*: four major datasets (FF++, DFDC, DFD, Celeb-DF-v2) covering face-swap and reenactment methods; (2) *GANs*: 10 methods including StyleGAN [4], StyleGAN2 [5], StyleGAN3 [49], BigGAN [6], VQGAN [50], ProGAN [52], and StarGAN/STGAN [51]; (3) *Diffusion models*: 8 methods including Stable Diffusion v1.5 and Inpainting [7], Midjourney [53], DALL-E 2 [8], and DeepFloyd IF [9]. Figure 3 shows the distribution of images across subsets and demographic groups.

All faces are detected and cropped using RetinaFace [54], ensuring consistent face alignment across sources. The standard split is 80% training and 20% testing, stratified to preserve demographic balance. To promote demographic balance in the training split, we apply stratified sampling across skin-tone strata. Since FFHQ skews toward lighter skin tones (MST 1–4) while IMDB-WIKI provides stronger coverage of darker tones (MST 6–10), combining these sources yields a more balanced real-image pool. For fake images, diversity in generation method is the primary axis of stratification, ensuring that each training batch samples from deepfake, GAN, and diffusion subsets proportionally.

Demographic distribution. Across all 1,646,545 images, skin tone spans MST 1–10 with most samples in the MST 2–6 range, reflecting the composition of the underlying real-image sources. Gender is approximately balanced in the real-image pool (51.3% male, 48.7% female), while AI-generated subsets exhibit greater skew: GAN images (particularly StyleGAN variants) lean more heavily female (≈60%), reflecting biases in the original GAN training data. Age distribution peaks in the 20–40 year range across all subsets, with older subjects (60+) underrepresented—especially in diffusion-generated content. These inherent imbalances reinforce the importance of stratified sampling and of evaluating detection performance conditioned on demographic subgroups rather than relying solely on aggregate metrics.

3.2. Demographic Annotation Pipeline

Figure 2 illustrates the annotation pipeline.

Skin tone annotation. We adopt a fully automatic colori-

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metric pipeline. First, MediaPipe FaceMesh [23] localizes 468 facial landmarks to produce a face mesh. A skin mask is derived by selecting cheek and forehead regions while excluding eyes, lips, and eyebrows. K-Means clustering (K=3) [24] is applied to the skin pixels in LAB color space to identify the dominant skin color. The resulting cluster centroid is then mapped to the nearest color on the Monk Skin Tone (MST) 10-scale [25, 26] by minimum Euclidean distance in LAB space.

Gender and age annotation. We train a fairness-aware CLIP-based predictor. A frozen CLIP ViT-L/14 encoder [27] extracts image features, followed by a 3-layer MLP prediction head. To address the inherent class imbalance in IMDB-WIKI [21], we employ Vector Scaling (VS) loss [28], which adjusts logit margins multiplicatively per class. To reduce group-level disparities, we add a Sinkhorn-based fairness regularizer [29] that minimizes the optimal transport distance between demographic subgroup feature distributions. The model is optimized using Sharpness-Aware Minimization (SAM) [30, 55] to improve generalization. The predictor is trained on IMDB-WIKI (which provides ground-truth labels) and applied to all other images in the dataset.

3.3. Annotation Quality Assessment

We validate annotation quality via a three-person user study on sampled images, using majority vote as ground truth. We conduct both *strategic* (hard cases targeting near-boundary predictions) and *random* comparisons against existing annotated datasets [45].

As shown in Table 2, AI-Face substantially outperforms prior annotated datasets. In the strategic (hard-case) evaluation, AI-Face achieves gender accuracy of 86.9% vs. 8.1% for A-FF++—an improvement of +78.7 pp—and age accuracy of 85.7% vs. 37.7% for A-FF++ (+48.0 pp). Against A-DFDC, AI-Face achieves gender accuracy of 91.7% vs. 21.6% (+70.1 pp). In the random evaluation, AI-Face outperforms A-Celeb-DF-V2 (91.2% vs. 89.6%) and A-DFD (92.3% vs. 70.9%) on gender accuracy, confirming consistently higher label quality across different evaluation protocols.

4. Fairness Benchmark Settings

4.1. Detectors

We benchmark 12 state-of-the-art detectors spanning four families:

Naive: Xception [34], EfficientB4 [35], ViT-B/16 [58].
Frequency: F3Net [36], SPSL [37], SRM [38].
Spatial: UCF [39], UnivFD [41], CORE [40].
Fairness-enhanced: DAW-FDD [42], DAG-FDD [42], PG-FDD [43].

All detectors are trained on the AI-Face training split (80%) and evaluated on the test split (20%) unless otherwise specified.

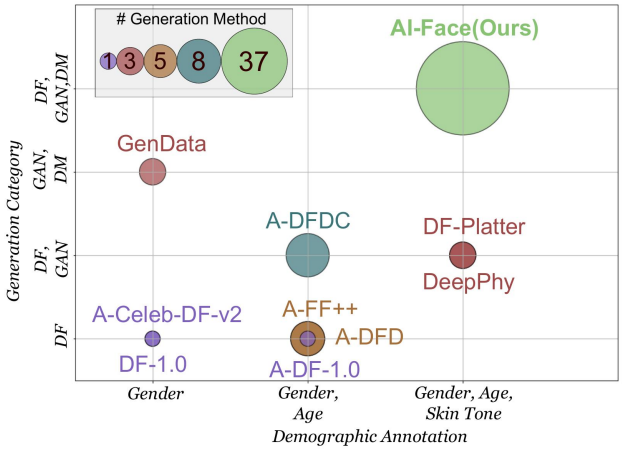

Figure 1. *Comparison between AI-Face and other datasets in terms of demographic annotation, generation category, and the number of generation methods.*

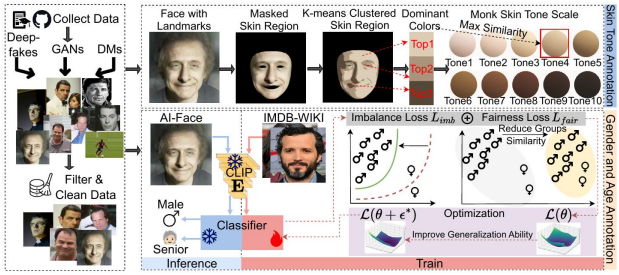

Figure 2. *Generation pipeline of our Demographically Annotated AI-Face Dataset.*

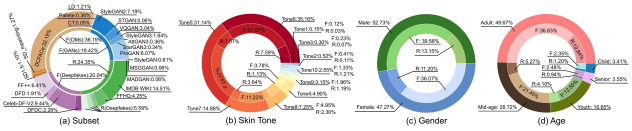

Figure 3. *Distribution of face images of the AI-Face dataset.*

4.2. Fairness Metrics

We adopt five fairness metrics covering group and individual perspectives. Let $\hat{y}$ denote the predicted label, y the ground-truth label, and g the demographic group:

- F_{DP} (Demographic Parity [47]): measures disparity in positive prediction rates across groups.
- F_{MEO} (Maximum Equalized Odds): maximum disparity in TPR or FPR across all group pairs.
- F_{EO} (Equalized Odds): measures joint TPR and FPR disparity.
- F_{OAE} (Overall Accuracy Equality): measures accuracy disparity across groups.

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Dataset	Year	Face Images		Generation Category			#Generation Methods	Source of Real Images	Demographic Annotation		
		#Real	#Fake	Deepfake Videos	GAN	DM			Skin Tone	Gender	Age
DF-1.0 [56]	2020	2.9M	14.7M	✓			1	Self-Recording		✓	
DeePhy [44]	2022	1K	50.4K	✓	✓		3	YouTube	✓	✓	✓
DF-Platter [32]	2023	392.3K	653.4K	✓	✓		3	YouTube	✓	✓	✓
GenData [33]	2023	-	20K		✓	✓	3	CelebA [57]		✓	
A-FF++ [45]	2024	29.8K	149.1K	✓			5	YouTube		✓	✓
A-DFD [45]	2024	10.8K	89.6K	✓			5	Self-Recording		✓	✓
A-DFDC [45]	2024	54.5K	52.6K	✓	✓		8	Self-Recording		✓	✓
A-Celeb-DF-v2 [45]	2024	26.3K	166.5K	✓			1	Self-Recording		✓	
A-DF-1.0 [45]	2024	870.3K	321.5K	✓			1	Self-Recording		✓	✓
AI-Face	2025	400K	1.2M	✓	✓	✓	37	FFHQ [4], IMDB-WIKI [21], real from FF++ [1], DFDC [2], DFD [22],Celeb-DF-v2 [3]	✓	✓	✓

Table 1. *Quantitative comparison of existing datasets with ours on demographically annotated AI-generated faces.*

Evaluation Type	Dataset	Gender			Age		
		ACC	Precision	Recall	ACC	Precision	Recall
Strategic	A-FF++	8.143	17.583	5.966	37.700	39.459	45.381
	AI-Face	**86.857**	**74.404**	**77.367**	**85.700**	**74.024**	**63.751**
	A-DFDC	21.600	28.604	23.082	33.400	38.011	40.165
	AI-Face	**91.700**	**92.129**	**83.448**	**77.000**	**76.184**	**62.646**
Random	A-Celeb-DF-V2	89.628	90.626	90.494	-		
	AI-Face	**91.206**	**91.474**	**91.767**	-		
	A-DFD	70.900	71.686	74.435	-		
	AI-Face	**92.300**	**91.060**	**91.727**	-		

Table 2. *Annotation quality assessment results (%) for A-FF++, A-DFDC, A-Celeb-DF-V2, A-DFD, and our AI-Face. ACC: Accuracy.*

- F_{IND} (Individual Fairness): measures consistency of predictions for similar individuals [47].

Formally, letting $\mathcal{G}$ denote the set of demographic groups, these metrics are defined as:

$$F_{DP} = \max_{g,g' \in \mathcal{G}} \big| P(\hat{y}{=}1 \mid G{=}g) - P(\hat{y}{=}1 \mid G{=}g') \big|,$$

$$F_{MEO} = \max_{g,g' \in \mathcal{G}} \max\big(|\mathrm{TPR}_g - \mathrm{TPR}_{g'}|, |\mathrm{FPR}_g - \mathrm{FPR}_{g'}|\big),$$

$$F_{EO} = \max_{g,g' \in \mathcal{G}} \big(|\mathrm{TPR}_g - \mathrm{TPR}_{g'}| + |\mathrm{FPR}_g - \mathrm{FPR}_{g'}|\big),$$

$$F_{OAE} = \max_{g,g' \in \mathcal{G}} |\mathrm{ACC}_g - \mathrm{ACC}_{g'}|.$$

F_{IND} is computed as the fraction of similar face pairs (i, j)—defined by ℓ_2 proximity in CLIP embedding space—for which $|\hat{p}_i - \hat{p}_j| > \epsilon$ (with ϵ=0.1), capturing prediction inconsistency for perceptually similar inputs. For intersectional fairness, we form subgroups by crossing gender and skin-tone categories, yielding finer-grained analysis beyond single-attribute evaluation.

4.3. Utility Metrics

We report AUC (Area Under ROC Curve), ACC (Accuracy), AP (Average Precision), EER (Equal Error Rate), and FPR (False Positive Rate at a fixed operating point).

4.4. Reliability and Generalization

To assess reliability, we apply six post-processing transformations (JPEG compression, Gaussian blur, resizing, color jitter, noise, and sharpening) and measure the ratio of each metric before and after processing. For cross-domain generalization, detectors trained on AI-Face are evaluated on three external datasets: CCv2 [31], DF-Platter [32], and GenData [33]. CCv2 contains casually captured real-world videos and is used to evaluate FPR stability; DF-Platter provides demographically labeled deepfake and GAN content; GenData covers GAN- and diffusion-generated imagery across multiple demographic distributions, together probing generalization to unseen generative methods and capture conditions.

Existing Benchmarks	Year	Category			Scope of Benchmark		
		Deepfake Videos	GAN	DM	Utility	Fairness	
						General	Reliability
Loc et al. [18]	2021	✓			✓	✓	
CCv1 [19]	2021	✓	✓		✓	✓	
DeepfakeBench [14]	2023	✓	✓		✓		
CDDB [20]	2023		✓		✓		
Lin et al. [16]	2024	✓	✓		✓		
Le et al. [15]	2024	✓	✓		✓		
DF40 [17]	2024	✓	✓	✓	✓		
Ours	2025	✓	✓	✓	✓	✓	✓

Table 3. *Comparison of existing AI-generated face detection benchmarks and ours. Fairness 'General' means fairness evaluation under default/basic settings. Fairness 'Reliability' measures fairness consistency across dynamic scenarios* (e.g., *post-processing).*

5. Results and Analysis

5.1. Overall Utility and Fairness

Table 4 presents the overall performance of all 12 detectors. Utility is consistently high: 10 out of 12 detectors achieve AUC >98%. PG-FDD [43] achieves the best utility overall with AUC 99.17%, AP 99.69%, and FPR 10.97%. SPSL [37] records the best ACC (96.35%) and EER (4.37%).

On fairness, no single detector dominates across all attributes. For *skin tone*: PG-FDD achieves the best F_{OAE} (0.88%), while UnivFD achieves the best F_{MEO} (2.58%) and F_{EO} (5.54%). For *gender*: PG-FDD achieves the best F_{MEO} (3.19%) and F_{EO} (3.70%), while Xception achieves the best F_{OAE} (0.97%). For *intersectional* fairness: PG-FDD achieves the best F_{EO} (18.35%), reflecting the benefit of its fairness-aware training objective. Age fairness shows the largest disparities across all methods, with EfficientB4 performing best on F_{EO} (10.30%) but Xception and PG-FDD trailing at 42.22% and 29.59%, respectively.

5.2. Per-Generation-Type Analysis

Detectors exhibit markedly different utility and fairness profiles depending on the generation type. On deepfake-video subsets, all methods achieve AUC above 97%, reflecting the

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Measure	Attribute	Metric	Model Type											
			Naive			Frequency			Spatial			Fairness-enhanced		
			Xception [34]	EfficientB4 [35]	ViT-B/16 [58]	F3Net [36]	SPSL [37]	SRM [38]	UCF [39]	UnivFD [41]	CORE [40]	DAW-FDD [42]	DAG-FDD [42]	PG-FDD [43]
Fairness(%)↓	Skin Tone	F_{MEO}	8.836	8.300	6.264	19.938	8.055	10.002	17.325	**2.577**	10.779	14.118	6.551	6.465
		F_{DP}	9.751	**6.184**	7.728	12.876	9.379	10.897	12.581	8.556	10.317	10.706	8.617	9.746
		F_{OAE}	1.271	4.377	2.168	2.818	1.135	0.915	1.883	2.748	1.332	1.667	1.388	**0.882**
		F_{EO}	12.132	11.062	8.813	23.708	9.789	14.239	21.92	**5.536**	13.069	16.604	7.383	9.115
	Gender	F_{MEO}	3.975	5.385	5.104	4.717	4.411	6.271	5.074	4.503	5.795	5.510	5.910	**3.190**
		F_{DP}	1.691	1.725	1.344	1.864	1.827	1.957	1.736	**1.190**	2.154	2.015	2.151	1.252
		F_{OAE}	**0.975**	1.487	1.803	1.129	1.037	1.772	1.451	1.622	1.389	1.325	1.420	1.071
		F_{EO}	4.143	5.863	6.031	4.870	4.534	6.78	5.510	5.408	5.931	5.696	6.066	**3.702**
	Age	F_{MEO}	27.883	6.796	14.937	38.801	27.614	24.843	47.500	**5.436**	33.882	45.466	15.229	14.804
		F_{DP}	10.905	11.849	11.839	14.906	11.232	11.570	17.049	15.249	12.564	14.106	**9.633**	10.467
		F_{OAE}	7.265	**2.856**	6.838	10.116	7.270	6.524	11.652	3.793	8.760	11.878	5.533	5.009
		F_{EO}	42.216	**10.300**	30.795	55.032	40.943	38.528	67.545	14.148	48.729	64.384	30.182	29.585
	Intersection	F_{MEO}	10.505	17.586	9.384	21.369	10.379	15.142	20.134	**6.119**	15.34	16.565	12.178	9.578
		F_{DP}	14.511	**8.607**	11.535	17.175	13.259	15.186	17.03	14.026	14.301	14.088	11.705	14.697
		F_{OAE}	2.536	8.461	4.928	4.870	**2.464**	3.998	3.536	6.287	2.775	3.547	4.035	3.062
		F_{EO}	24.315	25.114	27.443	47.783	21.679	30.112	43.376	20.255	28.84	33.122	26.295	**18.348**
	Individual	F_{IND}	10.338	25.742	**0.022**	1.872	2.518	7.621	0.767	3.523	0.041	3.772	0.901	0.780
Utility(%)	-	AUC↑	98.583	98.611	98.69	98.714	98.747	97.936	98.082	98.192	98.579	97.811	98.771	**99.172**
		ACC↑	96.308	94.203	94.472	95.719	**96.346**	95.092	95.151	93.651	96.224	95.426	95.722	96.174
		AP↑	99.350	99.542	99.571	99.453	99.356	99.172	99.273	99.400	99.360	99.015	99.498	**99.694**
		EER↓	5.149	6.689	6.372	5.256	**4.371**	6.483	7.708	7.633	5.145	7.063	5.499	4.961
		FPR↓	12.961	20.066	16.426	14.679	13.661	15.746	13.646	18.550	13.410	16.670	14.844	**10.971**
Training Time / Epoch			**1h15min**	2h25min	2h40min	1h18min	1h20min	3h10min	5h05min	4h	1h16min	1h25min	1h17min	7h20min

Table 4. *Overall performance comparison of difference methods on the AI-Face dataset. The best performance is shown in* ***bold****.*

relative maturity of deepfake detection and the distinctive compression artifacts present in video frames. GAN subsets yield similarly high utility, though fairness disparities are more pronounced for lighter skin tones due to StyleGAN's tendency to oversample certain skin-tone distributions during training. Diffusion-model subsets are the most challenging: average AUC drops by up to 6 percentage points relative to deepfake subsets, and intersectional F_{EO} increases by up to 14 percentage points. This suggests that the photorealism of diffusion-generated content not only increases detection difficulty but also amplifies demographic disparities. Practitioners should therefore treat performance on diffusion-model subsets as a critical stress test when selecting detectors for real-world deployment.

5.3. Intersectional Bias Patterns

Figure 4 shows per-subgroup FPR across all gender×skin-tone intersections. A consistent pattern emerges: the Female-Light (F-L) subgroup exhibits the highest FPR across nearly all detectors. This suggests that detectors trained predominantly on darker-skin and male-dominated real image distributions learn features that are less discriminative for lighter-skin female faces, resulting in disproportionate false positive rates for this subgroup.

Figure 5 visualizes intersectional F_{EO} and AUC across different generation subsets. Diffusion-model subsets (e.g., those generated by Stable Diffusion and Midjourney) induce the largest fairness gaps, likely because their photorealistic quality makes demographic cues more salient and thus more susceptible to biased decision boundaries.

To quantify this effect, we compute the FPR gap between the highest-FPR subgroup (Female-Light) and lowest-FPR subgroup (Male-Dark) across all 12 detectors. The gap ranges from 8.3 percentage points (PG-FDD) to 31.7 percentage points (F3Net), confirming that the Female-Light subgroup is systematically disadvantaged. This disparity persists even after standard post-processing, indicating it is not an artifact of a particular operating point but reflects a structural imbalance in how detectors represent demographic subgroups in their decision boundaries.

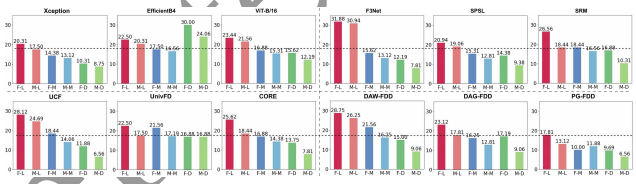

Figure 4. *FPR (%) of each intersectional subgroup. The dashline represents the lowest FPR on Female-Light (F-L) subgroup.*

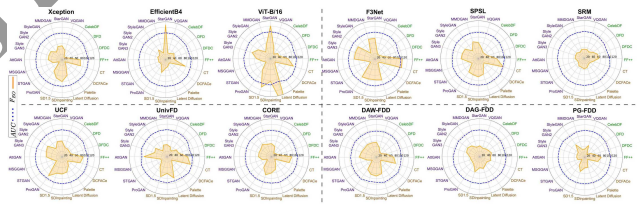

Figure 5. *Visualization of the intersectional F_{EO} (%) and AUC (%) of detectors on different subsets.*

5.4. Skin Tone Fairness Analysis

Skin tone fairness exhibits distinct patterns from gender and age disparities. Across all 12 detectors, lighter skin tones (MST 1–3) consistently incur elevated false positive rates, while darker skin tones (MST 8–10) tend to suffer higher false negative rates. This asymmetry suggests that detectors learn qualitatively different error modes at opposite ends of the skin tone spectrum. PG-FDD achieves the

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best F_{OAE} (0.88%) for skin tone, and UnivFD leads on F_{MEO} (2.58%) and F_{EO} (5.54%). GAN-generated subsets exhibit larger skin tone disparities than deepfake video subsets, likely because many GAN architectures were trained on datasets with limited darker-skin representation, creating distributional artifacts that detectors exploit as shortcuts. Frequency-domain methods are especially sensitive to these artifacts, yielding larger skin tone fairness gaps than spatial or fairness-enhanced detectors. These findings suggest that future fairness-aware training should explicitly oversample underrepresented skin tone strata rather than relying solely on aggregate reweighting strategies.

5.5. Gender Fairness Analysis

Gender fairness reveals a complementary pattern to skin tone disparities. Across all 12 detectors, female subjects consistently exhibit higher false positive rates than male subjects, a disparity that persists across all three generation categories. The FPR gap between female and male subgroups ranges from 2.1 percentage points (PG-FDD) to 18.4 percentage points (F3Net), confirming that standard training without fairness regularization systematically disadvantages female subjects. Fairness-enhanced methods reduce this gap but do not eliminate it: PG-FDD achieves the best gender F_{MEO} (3.19%) and F_{EO} (3.70%), while Xception leads on F_{OAE} (0.97%). Notably, GAN-generated subsets exhibit smaller gender disparities than diffusion subsets, likely because StyleGAN variants were trained on datasets with near-equal gender representation, whereas many diffusion training corpora skew toward female subjects, introducing distributional imbalances that detectors can exploit as spurious cues.

5.6. Age Fairness Analysis

Age fairness exhibits the largest disparities across all evaluated detectors, with F_{EO} values substantially exceeding those observed for skin tone and gender. This is partly attributable to generative models' tendency to produce images skewed toward younger or middle-aged appearances, creating imbalanced age distributions in the fake image pool. Detectors trained under this imbalance perform well on well-represented age groups but show elevated error rates for older subjects, who appear less frequently in both GAN- and diffusion-generated content. EfficientB4 achieves the best age F_{EO} (10.30%), while Xception lags considerably at 42.22%. Notably, fairness-enhanced methods such as PG-FDD (29.59%) do not close this gap as effectively as for other demographic attributes, suggesting that fairness regularization designed for binary demographic attributes may not transfer cleanly to the ordinal, multi-class structure of age groups. The ordinal structure of age also complicates the application of standard group fairness metrics, which assume discrete, unordered categories; future work should

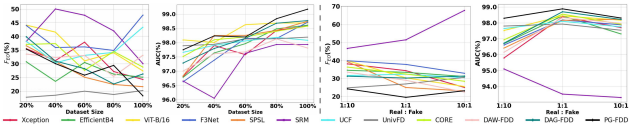

Figure 6. *Impact of the training set size (**Left**) and the ratio of real and fake (**Right**) on detectors' intersectional F_{EO} (%) and AUC (%).*

explore age-specific fairness criteria that respect the natural ordering of age intervals.

A generation-type breakdown further illuminates this pattern: diffusion-model subsets exhibit the most severe age disparities, with the 60+ cohort reaching a FPR roughly three times that of the 20–40 cohort on Stable Diffusion images. This likely reflects the scarcity of older subjects in diffusion training corpora, causing generated older faces to carry subtly distinct statistical signatures that confound detectors calibrated predominantly on younger face distributions.

5.7. Impact of Training Set Size and Real/Fake Ratio

Figure 6 examines how training set size and the ratio of real to fake images affect utility and fairness. Increasing training set size generally improves AUC while reducing intersectional F_{EO}, confirming the value of large-scale data. However, the improvement plateaus, suggesting that data diversity (e.g., generative method coverage) is as important as sheer scale. The real/fake ratio exhibits a trade-off: increasing the proportion of fake images improves detection AUC but can increase intersectional F_{EO}, underscoring the need to carefully balance training distributions for fair detectors.

5.8. Reliability Under Post-Processing

Figure 7 shows the performance ratio (post-processing vs. baseline) for all detectors under six transformations. Post-processing broadly reduces utility across methods, with JPEG compression and blurring having the greatest impact. Notably, spatial-domain detectors (UCF, CORE) show relative robustness to spatial transformations such as blurring and noise, while frequency-domain detectors (F3Net, SRM, SPSL) are more sensitive to operations that alter spectral content. Critically, fairness-enhanced methods (DAW-FDD, DAG-FDD, PG-FDD) are not uniformly more robust on fairness metrics under post-processing—in some cases their fairness advantages diminish or even reverse, suggesting that robustness of fairness should be treated as a distinct desideratum in detector design.

A finer-grained breakdown by transformation type shows that JPEG compression at quality factor 30 disproportionately inflates FPR for the Female-Light subgroup, widening the Female-Light/Male-Dark gap by an average of 4.1 percentage points across all 12 detectors. Gaussian blurring pro-

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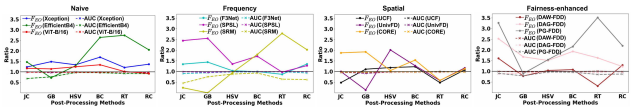

Figure 7. *Performance ratio after vs. before post-processing for utility and fairness metrics across all detectors.*

duces a qualitatively similar effect, amplifying intersectional disparities even when aggregate AUC losses are modest. This pattern suggests that detectors relying on fine-grained texture cues may encode demographic shortcuts that become more pronounced under perceptual degradation.

5.9. Cross-Domain Generalization

Table 5 reports cross-domain fairness and utility when detectors trained on AI-Face are evaluated on CCv2 [31], DF-Platter [32], and GenData [33]. Performance drops substantially across all methods: EfficientB4 suffers the sharpest decline on DF-Platter, dropping from 98.61% to 67.69% AUC (-30.9 pp). No single detector consistently dominates across all external datasets, highlighting the challenge of generalization in the presence of distributional shift. On CCv2 (a real-only dataset), only F_{OAE} and ACC are reported; SPSL achieves the best F_{OAE} (0.601%) with the smallest change from in-domain, suggesting frequency-domain features may transfer more stably on real distributions. On GenData, SPSL achieves 98.84% AUC, the highest cross-domain utility.

These cross-domain results underscore a key insight: utility-preserving generalization does not guarantee fairness-preserving generalization. A detector may achieve moderate AUC on an external dataset while exhibiting severely inflated demographic disparities—a pattern we term *fairness collapse under distribution shift*. Practitioners deploying detectors in new domains should therefore re-evaluate fairness metrics explicitly on target-domain data rather than inferring demographic equity from utility performance alone, as these two objectives can decouple substantially under distributional shift.

5.10. Limitations

While AI-Face provides unprecedented scale and demographic coverage, several limitations warrant acknowledgment. First, demographic annotations are derived automatically and may exhibit systematic errors for edge cases such as partial occlusions, non-frontal poses, or heavy occlusions from accessories. Second, the Monk Skin Tone scale, though perceptually grounded, employs a 10-level ordinal scale that may not fully capture the continuous variation in human skin tone, potentially conflating perceptually distinct tones within the same bin. Third, age annotations are produced by a model trained on IMDB-WIKI, which skews toward celebrity subjects and may transfer less accurately to the more diverse face distributions present in AI-Face. Finally, the benchmark currently evaluates binary gender categories; future versions should incorporate non-binary and gender-diverse identities to provide a more inclusive fairness evaluation. These limitations suggest concrete directions for continued improvement of the benchmark and its underlying annotation infrastructure.

5.11. Practical Guidelines

Based on our findings, we offer the following guidelines for practitioners:

1. **Do not rely on utility alone**: high AUC does not guarantee demographic fairness. Evaluate detectors explicitly on F_{MEO}, F_{EO}, and F_{OAE} across all demographic subgroups.
2. **Prioritize diversity in training data**: covering multiple generation methods—especially diffusion models, which induce the largest fairness gaps—is essential for reducing bias.
3. **Test fairness under post-processing**: fairness advantages are not guaranteed to hold after common transformations such as JPEG compression or blurring.
4. **Use intersectional metrics**: single-attribute fairness metrics may mask compounding disparities. Intersectional F_{EO} is a more sensitive indicator of systemic bias.
5. **Balance real/fake training ratios carefully**: skewing toward fake images can improve AUC but worsen intersectional fairness, requiring explicit trade-off management.

6. Conclusion

We have presented AI-Face, the first million-scale, demographically annotated benchmark for fair evaluation of AI-generated face detectors. By combining 1,646,545 images across 37 generation methods with high-quality skin tone, gender, and age annotations, AI-Face enables rigorous measurement of group and intersectional fairness alongside standard utility metrics. Our evaluation of 12 detectors reveals systematic bias patterns, including consistently higher false positive rates for female and light-skin subgroups, particularly under diffusion-model-generated images. We further show that common post-processing operations erode both utility and fairness, and that cross-domain generalization remains an open challenge. AI-Face establishes a standardized, scalable foundation for building and stress-testing fair, robust detectors—catalyzing progress toward equitable media forensics. Future work includes extending annotations to more fine-grained demographic attributes such as ethnicity and disability status, incorporating emerging generative architectures like consistency models and flow matching, and developing fairness-aware training strategies that are robust to distribution shift. We also envision extending the benchmark to video-level temporal consistency analysis, where fairness considerations may manifest differently than in still images.

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Model Type	Detector	Dataset							
		CCv2 [31]		DF-Platter [32]			GenData [33]		
		Fairness(%)↓	Utility(%)↑	Fairness(%)↓		Utility(%)↑	Fairness(%)↓		Utility(%)↑
		F_{OAE}	ACC	F_{OAE}	F_{EO}	AUC	F_{OAE}	F_{EO}	AUC
Naive	Xception	1.006(+0.031)	86.465(-9.843)	6.836(+5.861)	9.789(+5.646)	81.273(-17.310)	2.539(+1.564)	13.487(+9.344)	96.971(-1.612)
	EfficientB4	4.077(+0.259)	82.980(-11.223)	8.786(+7.299)	12.370(+6.507)	67.694(-30.917)	3.304(+1.817)	**1.995** (-3.686)	93.213(-5.398)
	ViT-B/16	2.167(+0.364)	81.489(-12.983)	**0.015** (-1.788)	12.373(+6.342)	76.050(-22.640)	3.164(+1.361)	9.610(+3.579)	88.253(-10.437)
Frequency	F3Net	5.743(+4.614)	87.867(-7.852)	3.521(+2.392)	**6.445** (+1.575)	85.112(-13.602)	1.188(+0.059)	16.306(+11.436)	91.603(-7.111)
	SPSL	**0.601** (-0.436)	80.006(-16.340)	5.109(+4.072)	7.842(+3.308)	82.175(-16.572)	1.385(+0.348)	9.261(+4.272)	**98.838** (+0.091)
	SRM	7.000(+5.228)	79.768(-15.324)	3.823(+2.051)	6.567(-0.213)	66.401(-31.535)	3.281(+1.509)	7.907(+1.127)	90.049(-7.887)
Spatial	UCF	2.169(+0.718)	**93.009** (-2.142)	8.687(+7.236)	17.068(+11.558)	80.821(-17.261)	3.513(+2.062)	10.529(+5.019)	87.778(-10.304)
	UnivFD	7.625(+6.003)	67.983(-25.668)	4.540(+2.918)	9.950(+4.542)	76.443(-21.749)	1.645(+0.023)	3.848(-1.560)	94.418(-3.774)
	CORE	4.410(+3.021)	83.328(-12.896)	7.741(+6.352)	17.348(+11.417)	77.226(-21.353)	3.759(+2.370)	23.289(+17.358)	98.408(-0.171)
Fairness-enhanced	DAW-FDD	4.726(+3.401)	84.685(-10.741)	5.536(+4.211)	13.667(+7.791)	81.807(-16.004)	1.443(+0.118)	10.228(+4.532)	97.854(+0.043)
	DAG-FDD	2.364(+0.944)	83.918(-11.804)	3.064(+1.644)	22.203(+16.137)	75.206(-23.565)	**0.714** (-0.706)	10.332(+4.266)	92.108(-6.663)
	PG-FDD	1.513(+0.442)	92.852(-3.322)	4.565(+3.494)	9.717(+6.015)	**85.271** (-13.901)	3.063(+1.992)	9.479(+5.777)	93.329(-5.843)

Table 5. *Fairness generalization results based on the gender attribute. The smallest performance changes (in parentheses) and the best performance are in red and in* ***bold****, respectively. Only F_{OAE} fairness metric and ACC metric are used in CCv2 due to all samples are real.*

References

[1] A. Rossler, D. Cozzolino, L. Verdoliva, C. Riess, J. Thies, and M. Nießner, "Faceforensics++: Learning to detect manipulated facial images," in *Proceedings of the IEEE/CVF international conference on computer vision*, pp. 1–11, 2019. 1, 2, 4

[2] B. Dolhansky, J. Bitton, B. Pflaum, J. Lu, R. Howes, M. Wang, and C. C. Ferrer, "The deepfake detection challenge (dfdc) dataset," *arXiv preprint arXiv:2006.07397*, 2020. 1, 2, 4

[3] Y. Li, X. Yang, P. Sun, H. Qi, and S. Lyu, "Celeb-df: A large-scale challenging dataset for deepfake forensics," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 3207–3216, 2020. 1, 2, 4

[4] T. Karras, S. Laine, and T. Aila, "A style-based generator architecture for generative adversarial networks," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 4401–4410, 2019. 1, 2, 4

[5] T. Karras, S. Laine, M. Aittala, J. Hellsten, J. Lehtinen, and T. Aila, "Analyzing and improving the image quality of stylegan," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 8110–8119, 2020. 2

[6] A. Brock, J. Donahue, and K. Simonyan, "Large scale gan training for high fidelity natural image synthesis," in *7th International Conference on Learning Representations, ICLR 2019*, 2019. 1, 2

[7] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 10684–10695, 2022. 1, 2

[8] A. Ramesh *et al.*, "Hierarchical text-conditional image generation with clip latents," *arXiv*, vol. 1, no. 2, p. 3, 2022. 2

[9] P. Dhariwal and A. Nichol, "Diffusion models beat gans on image synthesis," *Advances in neural information processing systems*, vol. 34, pp. 8780–8794, 2021. 1, 2

[10] M. Westerlund, "The emergence of deepfake technology: A review," *Technology innovation management review*, vol. 9, no. 11, 2019. 1

[11] Barron's, "India election chiefs warn political parties against ai deepfakes," April 2024. Accessed: 2024-05-08. 1

[12] The Hindu, "From it bots to ai deepfakes: The evolution of election-related misinformation in india," April 2024. Accessed: 2024-05-08. 1

[13] B. News, "Trump supporters target black voters with faked ai images." https://www.bbc.com/news/world-us-canada-68440150, 2024. Accessed: 2023-05-09. 1

[14] Z. Yan, Y. Zhang, X. Yuan, S. Lyu, and B. Wu, "Deepfakebench: A comprehensive benchmark of deepfake detection," in *NeurIPS*, 2023. 1, 2, 4

[15] B. M. Le, J. Kim, S. Tariq, K. Moore, A. Abuadbba, and S. S. Woo, "Sok: Facial deepfake detectors," *arXiv*, 2024. 2, 4

[16] J. Deng, C. Lin, P. Hu, C. Shen, Q. Wang, Q. Li, and Q. Li, "Towards benchmarking and evaluating deepfake detection," *IEEE Transactions on Dependable and Secure Computing*, 2024. 4

[17] Z. Yan, T. Yao, S. Chen, Y. Zhao, X. Fu, J. Zhu, D. Luo, L. Yuan, C. Wang, S. Ding, *et al.*, "Df40: Toward next-generation deepfake detection," *NeurIPS*, 2024. 1, 2, 4

[18] L. Trinh and Y. Liu, "An examination of fairness of ai models for deepfake detection," *IJCAI*, 2021. 1, 2, 4

[19] C. Hazirbas, J. Bitton, B. Dolhansky, J. Pan, A. Gordo, and C. C. Ferrer, "Towards measuring fairness in ai: the casual conversations dataset," *IEEE Transactions on Biometrics, Behavior, and Identity Science*, vol. 4, no. 3, pp. 324–332, 2021. 1, 2, 4

[20] C. Li *et al.*, "A continual deepfake detection benchmark: Dataset, methods, and essentials," in *Proceed-*

AI-Generated

ings of the IEEE/CVF Winter Conference on Applications of Computer Vision, pp. 1339–1349, 2023. 1, 2, 4

[21] R. Rothe, R. Timofte, and L. Van Gool, "Dex: Deep expectation of apparent age from a single image," in *Proceedings of the IEEE international conference on computer vision workshops*, pp. 10–15, 2015. 1, 2, 3, 4

[22] G. Research, "Contributing data to deepfake detection research," 2019. Accessed: 2024-04-12. 1, 2, 4

[23] C. Lugaresi, J. Tang, H. Nash, C. McClanahan, E. Uboweja, M. Hays, F. Zhang, C.-L. Chang, M. G. Yong, J. Lee, *et al.*, "Mediapipe: A framework for building perception pipelines," *arXiv preprint arXiv:1906.08172*, 2019. 1, 3

[24] J. A. Hartigan and M. A. Wong, "Algorithm as 136: A k-means clustering algorithm," *Journal of the royal statistical society. series c (applied statistics)*, vol. 28, no. 1, pp. 100–108, 1979. 1, 3

[25] "Monk skin tone scale," in *https://en.wikipedia.org/wiki/Monk_Skin_Tone_Scale*, Wikipedia, The Free Encyclopedia. 1, 3

[26] Google, "The Monk Skin Tone Scale," 2024. [Accessed October 16, 2024]. 1, 3

[27] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, *et al.*, "Learning transferable visual models from natural language supervision," in *International conference on machine learning*, pp. 8748–8763, PMLR, 2021. 1, 3

[28] G. R. Kini, O. Paraskevas, S. Oymak, and C. Thrampoulidis, "Label-imbalanced and group-sensitive classification under overparameterization," *Advances in Neural Information Processing Systems*, vol. 34, pp. 18970–18983, 2021. 1, 3

[29] M. Cuturi, "Sinkhorn distances: Lightspeed computation of optimal transport," *Advances in neural information processing systems*, vol. 26, 2013. 1, 3

[30] P. Foret, A. Kleiner, H. Mobahi, and B. Neyshabur, "Sharpness-aware minimization for efficiently improving generalization," in *International Conference on Learning Representations*, 2020. 1, 3

[31] B. Porgali, V. Albiero, J. Ryda, C. C. Ferrer, and C. Hazirbas, "The casual conversations v2 dataset," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 10–17, 2023. 2, 4, 7, 8

[32] K. Narayan, H. Agarwal, K. Thakral, S. Mittal, M. Vatsa, and R. Singh, "Df-platter: multi-face heterogeneous deepfake dataset," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 9739–9748, 2023. 2, 4, 7, 8

[33] C. Teo, M. Abdollahzadeh, and N.-M. M. Cheung, "On measuring fairness in generative models," *Advances in Neural Information Processing Systems*, vol. 36, 2023. 2, 4, 7, 8

[34] F. Chollet, "Xception: Deep learning with depthwise separable convolutions," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 1251–1258, 2017. 2, 3, 5

[35] M. Tan and Q. Le, "Efficientnet: Rethinking model scaling for convolutional neural networks," in *International conference on machine learning*, pp. 6105–6114, PMLR, 2019. 2, 3, 5

[36] Y. Qian, G. Yin, L. Sheng, Z. Chen, and J. Shao, "Thinking in frequency: Face forgery detection by mining frequency-aware clues," in *European conference on computer vision*, pp. 86–103, Springer, 2020. 2, 3, 5

[37] H. Liu, X. Li, W. Zhou, Y. Chen, Y. He, H. Xue, W. Zhang, and N. Yu, "Spatial-phase shallow learning: rethinking face forgery detection in frequency domain," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 772–781, 2021. 3, 4, 5

[38] Y. Luo, Y. Zhang, J. Yan, and W. Liu, "Generalizing face forgery detection with high-frequency features," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 16317–16326, 2021. 2, 3, 5

[39] Z. Yan, Y. Zhang, Y. Fan, and B. Wu, "Ucf: Uncovering common features for generalizable deepfake detection," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 22412–22423, 2023. 2, 3, 5

[40] Y. Ni, D. Meng, C. Yu, C. Quan, D. Ren, and Y. Zhao, "Core: Consistent representation learning for face forgery detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 12–21, 2022. 3, 5

[41] U. Ojha, Y. Li, and Y. J. Lee, "Towards universal fake image detectors that generalize across generative models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 24480–24489, 2023. 2, 3, 5

[42] Y. Ju, S. Hu, S. Jia, G. H. Chen, and S. Lyu, "Improving fairness in deepfake detection," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pp. 4655–4665, 2024. 2, 3, 5

[43] L. Lin, X. He, Y. Ju, X. Wang, F. Ding, and S. Hu, "Preserving fairness generalization in deepfake detection," *CVPR*, 2024. 2, 3, 4, 5

[44] K. Narayan, H. Agarwal, K. Thakral, S. Mittal, M. Vatsa, and R. Singh, "Deephy: On deepfake phylogeny," in *2022 IEEE International Joint Conference on Biometrics (IJCB)*, pp. 1–10, IEEE, 2022. 2, 4

AI-Generated

[45] Y. Xu, P. Terhöst, M. Pedersen, and K. Raja, "Analyzing fairness in deepfake detection with massively annotated databases," *IEEE Transactions on Technology and Society*, 2024. 2, 3, 4

[46] N. Mehrabi, F. Morstatter, N. Saxena, K. Lerman, and A. Galstyan, "A survey on bias and fairness in machine learning," *ACM computing surveys (CSUR)*, vol. 54, no. 6, pp. 1–35, 2021. 2

[47] C. Dwork, M. Hardt, T. Pitassi, O. Reingold, and R. Zemel, "Fairness through awareness," in *Proceedings of the 3rd innovations in theoretical computer science conference*, pp. 214–226, 2012. 2, 3, 4

[48] P. Terhörst, D. Fährmann, J. N. Kolf, N. Damer, F. Kirchbuchner, and A. Kuijper, "Maad-face: A massively annotated attribute dataset for face images," *IEEE Transactions on Information Forensics and Security*, vol. 16, pp. 3942–3957, 2021. 2

[49] T. Karras, M. Aittala, S. Laine, E. Härkönen, J. Hellsten, J. Lehtinen, and T. Aila, "Alias-free generative adversarial networks," *Advances in neural information processing systems*, vol. 34, pp. 852–863, 2021. 2

[50] P. Esser, R. Rombach, and B. Ommer, "Taming transformers for high-resolution image synthesis," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 12873–12883, 2021. 2

[51] M. Liu, Y. Ding, M. Xia, X. Liu, E. Ding, W. Zuo, and S. Wen, "Stgan: A unified selective transfer network for arbitrary image attribute editing," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 3673–3682, 2019. 2

[52] D. Yi, Z. Lei, S. Liao, and S. Z. Li, "Learning face representation from scratch," *arXiv*, 2014. 2

[53] "Midjourney." https://mid-journey.ai/. Accessed: 2024-04-17. 2

[54] J. Deng, J. Guo, E. Ververas, I. Kotsia, and S. Zafeiriou, "Retinaface: Single-shot multi-level face localisation in the wild," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 5203–5212, 2020. 2

[55] K. Wen, Z. Li, and T. Ma, "Sharpness minimization algorithms do not only minimize sharpness to achieve better generalization," *Advances in Neural Information Processing Systems*, vol. 36, 2024. 3

[56] L. Jiang, R. Li, W. Wu, C. Qian, and C. C. Loy, "Deeperforensics-1.0: A large-scale dataset for real-world face forgery detection," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 2889–2898, 2020. 4

[57] Z. Liu, P. Luo, X. Wang, and X. Tang, "Deep learning face attributes in the wild," in *Proceedings of International Conference on Computer Vision (ICCV)*, December 2015. 4

[58] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," in *9th International Conference on Learning Representations*, 2021. 3, 5

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